



DUBLIN CITY UNIVERSITY

AUGUST/RESIT EXAMINATIONS 2013/2014

MODULE: CA208 - Logic

PROGRAMME(S): CASE BSc. in Computer Applications (Sft. Eng.)

YEAR: 2

EXAMINERS: Dr. David Sinclair, Ext no. 5510.

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer 10 Questions.
Answer all 8 Questions in Section A.
Answer any 2 questions from Section B.
All questions carry equal marks.

Please do not turn over this page until instructed to do so

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones

Section A

QUESTION 1

[TOTAL MARKS: 10]

[10 Marks]

Simplify the following propositional formulae:

1. $A \wedge A$
2. $A \vee A$
3. $A \wedge \neg A$
4. $A \vee \neg A$
5. $A \rightarrow \text{false}$

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 10]

[10 Marks]

Show that

$$(A \rightarrow C) \wedge (B \rightarrow C) \equiv (A \vee B) \rightarrow C$$

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 10]

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$\neg(p \wedge q) \leftrightarrow (\neg p \vee \neg q)$$

[End of Question 3]

QUESTION 4

[TOTAL MARKS: 10]

[10 Marks]

Using the Gentzen System $\mathcal{G}$ prove

$$\vdash (A \vee (B \wedge C)) \rightarrow ((A \vee B) \wedge (A \vee C))$$

[End of Question 4]

QUESTION 5**[TOTAL MARKS: 10]****[10 Marks]**

Using the Hilbert System $\mathcal{H}$ and the following derived rules and theorem,

$$\text{Deduction: } \frac{U \cup \{A\} \vdash B}{U \vdash A \rightarrow B}$$

prove

$$\vdash (A \rightarrow (B \rightarrow C)) \rightarrow ()$$

[End of Question 5]

QUESTION 6**[TOTAL MARKS: 10]****[10 Marks]**

Convert the following propositional formula into Conjunctive Normal Form and using resolution show that this formula is unsatisfiable.

$$(p \vee q) \wedge (q \rightarrow r) \wedge \neg r \wedge (p \rightarrow q)$$

[End of Question 6]

QUESTION 7**[TOTAL MARKS: 10]****[10 Marks]**

Using Gentzen System $\mathcal{G}$ prove

$$\vdash \forall x.(A \rightarrow B(x)) \rightarrow (A \rightarrow \forall x.B(x))$$

[End of Question 7]

QUESTION 8**[TOTAL MARKS: 10]****[10 Marks]**

Using General Resolution show that the following formula in clausal form is unsatisfiable.

$$\{\{\neg R(a, a)\}, \{R(x, f(x))\}, \{\neg R(x, y), \neg R(y, z), R(x, z)\}, \{\neg R(x, y), R(y, x)\}\}$$

[End of Question 8]

Section B

QUESTION 9**[TOTAL MARKS: 10]****[10 Marks]**

Given the relation `road(A,B,N)` is true if a one-way road of length `N` km exists from `A` to `B`. Assume that the set roads does not contain any loops where it is possible to travel from any town back to the same town, write the following Prolog relations:

- `routen(X,Y,N)` is true if there exists a route of length `N`, with zero or more intermediate towns, between `X` and `Y`.
- `canwalk(X,Y)` is true if there is a route between `X` and `Y` and the total distance is less than or equal to 4 km.

[End of Question 9]

QUESTION 10**[TOTAL MARKS: 10]****[10 Marks]**

Write the following Prolog relations without using any built-in Prolog list manipulation predicates.

- `inlist(A,B)` is true if the element `A` is contained in list `B`.
- `myappend(A,B,C)` is true if list `C` is list `B` appended to list `A`.
- `mydelete(A,B,C)` is true if list `C` is list `B` with the element `A` removed.

[End of Question 10]

QUESTION 11**[TOTAL MARKS: 10]****[10 Marks]**

Describe what *negation as failure* means in Prolog. Why is negation problematical in Prolog? Illustrate your answer with examples.

[End of Question 11]

Appendix

Semantic Tableaux

α	α_1	α_2	β	β_1	β_2
$\neg\neg A_1$	A_1		$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \wedge A_2$	A_1	A_2	$B_1 \vee B_2$	B_1	B_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$	$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$	$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2	$\neg(B_1 \downarrow B_2)$	B_1	B_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$	$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$	$\exists x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$	$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2	β	β_1	β_2
$\neg\neg A$	A		$B_1 \wedge B_2$	B_1	B_2
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \vee A_2$	A_1	A_2	$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$A_1 \rightarrow A_2$	$\neg A_1$	A_2	$\neg(B_1 \uparrow B_2)$	B_1	B_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \downarrow B_2)$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2	$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$	$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$	$\forall x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$	$\neg\exists x.A(x)$	$\neg A(a)$

Hilbert Logic

The *Hilbert System* $\mathcal{H}$ for propositional logic has 3 axioms,

Axiom 1: $\vdash (A \rightarrow (B \rightarrow A))$

Axiom 2: $\vdash (A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$

Axiom 3: $\vdash (\neg B \rightarrow \neg A) \rightarrow (A \rightarrow B)$

and only 1 rule of inference.

Modus Ponens:
$$\frac{\vdash A \quad \vdash A \rightarrow B}{\vdash B}$$

The *Hilbert System* $\mathcal{H}$ for first-order logic has 2 additional axioms,

Axiom 4: $\vdash \forall x.A(x) \rightarrow A(a)$ x is a free variable in $A(x)$

Axiom 5: $\vdash \forall x.(A \rightarrow B(x)) \rightarrow (A \rightarrow \forall x.B(x))$ x is a free variable in $B(x)$
 x is not free in $A(x)$

and 1 additional rule of inference.

Generalisation:
$$\frac{\vdash A(a)}{\vdash \forall x.A(a)}$$

[END OF EXAM]